WORKED SOLUTION

STEP I 2007, Question 9

Mechanics • Friction

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Read the question before you go on. It is in the STEP Support Programme's question database:

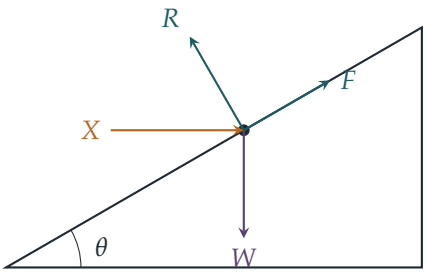
step.maths.org/questions/07-s1-q9

The key idea

This is two cases of limiting friction, which differ only in the direction of the friction force. In each case we resolve along and perpendicular to the plane and eliminate the normal reaction, which gives each horizontal force as W times a ratio of trigonometric expressions. Dividing one by the other removes W and leaves a relation between k, μ and θ alone. The inequality then follows from the fact that $\sin\theta\cos\theta = \frac{1}{2}\sin 2\theta$ can never exceed $\frac{1}{2}$.

Solution

We model the body as a particle and assume friction is limiting, $F = \mu R$, in both cases. Write $s = \sin\theta$ and $c = \cos\theta$ to keep the algebra compact.



The horizontal force pushes the particle into the plane, so it has a component Xc up the slope and a component Xs into the plane. The weight has a component Ws down the slope and Wc into the plane.

About to slide down: force X

The particle is on the point of moving down, so friction μR acts *up* the plane (as drawn). Resolving perpendicular to the plane, then along it,

$$R = Wc + Xs, \qquad Xc + \mu R = Ws.$$

Substituting for R,

$$Xc + \mu Wc + \mu Xs = Ws \quad \Longrightarrow \quad X(c + \mu s) = W(s - \mu c). \tag{1}$$

About to slide up: force kX

Now the particle is on the point of moving up, so friction acts *down* the plane. Similarly,

$$R' = Wc + kXs, \qquad kXc = Ws + \mu R',$$

so

$$kX(c - \mu s) = W(s + \mu c). \tag{2}$$

? **Ask yourself** Are we allowed to divide by everything in sight here?

Yes, given the physical set-up. A positive force X is *needed* to stop the particle sliding down, so the particle would slide without it, which means $\tan\theta > \mu$, i.e. $s - \mu c > 0$. Then (1) gives $X > 0$, and $c + \mu s > 0$ anyway. For a horizontal push to be able to drive the particle up the slope at all, (2) needs $c - \mu s > 0$, i.e. $\tan\theta < 1/\mu$. Together these force $\mu < \tan\theta < 1/\mu$, and in particular $\mu < 1$; we shall need this at the end.

The relation

Dividing (2) by (1) removes both X and W:

$$k\,\frac{c - \mu s}{c + \mu s} = \frac{s + \mu c}{s - \mu c} \quad \Longrightarrow \quad k(c - \mu s)(s - \mu c) = (s + \mu c)(c + \mu s).$$

Expanding each side and using $s^2 + c^2 = 1$:

$$(c - \mu s)(s - \mu c) = sc - \mu c^2 - \mu s^2 + \mu^2 sc = (1 + \mu^2)sc - \mu,$$
$$(s + \mu c)(c + \mu s) = sc + \mu s^2 + \mu c^2 + \mu^2 sc = (1 + \mu^2)sc + \mu.$$

So $k(1 + \mu^2)sc - k\mu = (1 + \mu^2)sc + \mu$, which rearranges to

$$(k - 1)(1 + \mu^2)\sin\theta\cos\theta = \mu(k + 1),$$

as required.

The inequality

The right-hand side is positive (since $\mu > 0$ and $k > 0$), and so is $(1 + \mu^2)\sin\theta\cos\theta$ for an acute θ; hence $k > 1$. (This is no surprise: pushing the particle up takes more than holding it.) Now

$$\sin\theta\cos\theta = \tfrac{1}{2}\sin 2\theta \leq \tfrac{1}{2},$$

and since $(k-1)(1+\mu^2) > 0$ we may multiply this inequality by it:

$$\mu(k+1) = (k-1)(1+\mu^2)\sin\theta\cos\theta \leq \tfrac{1}{2}(k-1)(1+\mu^2).$$

Doubling and collecting the terms in k:

$$2\mu k + 2\mu \leq k(1+\mu^2) - (1+\mu^2) \quad \Longrightarrow \quad k(1-2\mu+\mu^2) \geq 1 + 2\mu + \mu^2,$$

that is, $k(1-\mu)^2 \geq (1+\mu)^2$. We showed above that $\mu < 1$, so $(1-\mu)^2 > 0$ and we can divide by it without changing the direction:

$$k \geq \frac{(1+\mu)^2}{(1-\mu)^2}.$$

Equality holds exactly when $\sin 2\theta = 1$, i.e. $\theta = 45°$.

Ask yourself Why must we know $\mu < 1$ before the last step?

If $\mu = 1$ we would be dividing by zero (and the line before would read $0 \geq 4$, which is false); if $\mu > 1$ dividing would still be legal, but the conditions on θ would already have failed. Physically, with $\mu \geq 1$ the two requirements $\tan\theta > \mu \geq 1$ and $\tan\theta < 1/\mu \leq 1$ contradict each other: either the particle does not need holding up at all, or no horizontal push can ever start it up the slope. So the question implicitly has $\mu < 1$.

Checks

- Take $\mu = \frac{1}{2}$ and $\theta = 45°$, so $s = c = 1/\sqrt{2}$. From (1) and (2), $X = W \cdot \frac{1/2}{3/2} = \frac{1}{3}W$ and $kX = W \cdot \frac{3/2}{1/2} = 3W$, so $k = 9 = (1.5/0.5)^2$: the equality case. With $\theta = 60°$ instead, the same formulae give $k \approx 25.2 > 9$.
- If $\mu \to 0$ then $X = kX = W\tan\theta$ and $k \to 1$, matching both the identity (both sides 0) and the bound 1.
- A numerical check of the identity and the inequality over 10^5 random choices of $\mu \in (0,1)$ and admissible θ found no failures.

Common slips

- Taking $R = W\cos\theta$: the horizontal force also presses the particle into the plane and adds $X\sin\theta$ to the reaction.
- Keeping friction in the same direction in both cases. It always opposes the motion that is about to happen.
- Dividing by $(1-\mu)^2$ without saying why it is positive (and non-zero).

Beyond STEP

A neat way to see this is through the *angle of friction* λ, with $\tan\lambda = \mu$: the normal reaction and limiting friction combine into a single force at angle λ to the normal, and (1) and (2) become $X = W\tan(\theta - \lambda)$ and $kX = W\tan(\theta + \lambda)$. So $k = \tan(\theta + \lambda)/\tan(\theta - \lambda)$, and the bound is the statement that this ratio is smallest at $\theta = 45^\circ$. Friction-angle arguments of this kind, and the idea of a "friction cone", reappear in university statics and in the theory of contact. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats friction and force analysis with the care first-year courses expect.

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